

CUHK(SZ)-CSC1003 Midterm Exam

Oct. 21, 2024

Name:

Student ID:

Note:

- This is a closed exam. No notes, calculators, or dictionaries are allowed.
- This exam paper has six pages (including cover page), in single-sided printing.
- Answer all questions within the time limit of 100 minutes.

Example Question:

Read the following Java program, and write down your answer(s).

```
1 public class HelloWorld {
2     public static void main(String[] args) {
3         System.out.println("Hello World!");
4     }
5 }
```

What is the output of the program when running “java HelloWorld”?

Hello World!

[illegible]

1. (10 points) Read the following Java program, and write down your answer(s).

```
1 public class Q1 {  
2     public static void main(String[] args) {  
3         int s = 0;  
4         for (int i = 0; i < args.length; i++) s += Integer.parseInt(args[i]);  
5         int a = s / args.length;  
6         System.out.println("a = " + a);  
7     }  
8 }
```

What is the command to compile the program?

“javac” or “javac Q1.java”

What is the output of the program when running “java Q1 1 2 3 4”?

a = 2

2. (10 points) Read the following Java program, and write down your answer(s).

```
1 public class Q2 {  
2     public static void main(String[] args) {  
3         int[] array = new int[args.length];  
4         int start = 0;  
5         int end = args.length - 1;  
6         while (start < end) {  
7             int temp = Integer.parseInt(args[start]);  
8             array[start] = Integer.parseInt(args[end]);  
9             array[end] = temp;  
10            start++;  
11            end--;  
12        }  
13        for (int i = 0; i < array.length; i++) {  
14            System.out.print(array[i] + " ");  
15        }  
16    }  
17 }
```

What is the output of the program when running “java Q2 1 2 3 4 5 > result.txt”? Here “>” denotes the standard output redirection.

“5 4 0 2 1” in a file named “result.txt”

If the program is to reverse the command-line input numbers, then there is a mistake in the program. Point out the line number of the mistake, and correct it.

Line 6, “start < end” to “start <= end”

3. (10 points) Read the following Java program, and write down your answer(s).

```
1 public class Q3 {  
2     public static void main(String args[]){  
3         int r, sum = 0, temp;  
4         int n = Integer.parseInt(args[0]);  
5         temp = n;  
6         while (n > 0){  
7             r = n % 10;  
8             sum = sum * 10 + r;  
9             n = n / 10;  
10        }  
11        if(temp == sum) {  
12            System.out.println("true");  
13        } else {  
14            System.out.println("false");  
15        }  
16    }  
17 }
```

(Hint: “%” calculates the remainder of integer divisions, i.e., $9 \% 3$ is 0, $8 \% 3$ is 2, ...)
What is the output of the program when running “java Q3 121212”?

false

What is the output of the program when running “java Q3 121121”?

true

4. (10 points) Read the following Java program, and write down your answer(s).

```
1 public class Q4 {  
2     public static void main(String[] args) {  
3         int x = Integer.parseInt(args[0]);  
4         int y = Integer.parseInt(args[1]);  
5         int g = 1;  
6         for(int i = 1; i <= x && i <= y; i++) {  
7             if ((x % i) == 0 && (y % i) == 0) g = i;  
8         }  
9         System.out.println("g = " + g);  
10    }  
11 }
```

What is the output of the program when running “java Q4 24 84”?

g = 12

5. (10 points) Read the following Java program, and write down your answer(s).

```
1 public class Q5 {  
2     public static void main(String[] args) {  
3         int[] a = {1, 2, 3, 4, 5}; int[] b = {4, 3, 2, 1};  
4         System.out.println("a.length = " + a.length);  
5         System.out.println("b.length = " + b.length);  
6         b = a;  
7         for (int i = 0; i < b.length; i++) b[i]++;  
8         System.out.print("Array a:");  
9         for (int i = 0; i < a.length; i++) System.out.print(" " + a[i]);  
10        System.out.println();  
11        System.out.print("Array b:");  
12        for (int i = 0; i < b.length; i++) System.out.print(" " + b[i]);  
13    }  
14 }
```

What is the output of the program when running “java Q5”?

a.length = 5 b.length = 4 Array a: 2 3 4 5 6 Array b: 2 3 4 5 6

6. (10 points) Read the following Java program, and write down your answer(s).

```
1 public class Q6 {  
2     public static void main(String[] args) {  
3         int[] a = new int[args.length];  
4         for (int i = 0; i < args.length; i++)  
5             a[i] = Integer.parseInt(args[i]);  
6         for (int i = 0; i < a.length; i++)  
7             for (int j = i+1; j < a.length; j++)  
8                 if (a[i] > a[j]) {  
9                     int temp = a[i]; a[i] = a[j]; a[j] = temp;  
10                }  
11        for (int i = 0; i < a.length; i++) System.out.print(a[i] + " ");  
12    }  
13 }
```

What is the output of the program when running “java Q6 3 8 9 5 2 4 1”?

1 2 3 4 5 8 9

7. (10 points) Consider a program, which takes a positive integer n as input and prints the binary representation of n , by casting out powers of 2 in decreasing order. For example, the expected output for input “19” is “10011” because $19 = 1 \times 2^4 + 1 \times 2^1 + 1 \times 2^0$.

```
1 public class Q7 {
2     public static void main(String[] args) {
3         int n = Integer.parseInt(args[0]);
4         int power = 1;
5         while (power < n/2) power *= 2;
6         while (power > 0) {
7             if (n < power) {
8                 System.out.print(0);
9             } else {
10                System.out.print(1);
11                n -= power;
12            }
13            power /= 2;
14        }
15    }
16 }
```

The program has one mistake. Point out the line number of the mistake and correct it.

Line 5, “power < n/2” to “power <= n/2”

8. (15 points) Study the following Java program. The program reads an array of random integer numbers (each number is between 0 and 255), and then outputs the frequencies (denoted by “counts”) of each number in the array.

```
1 public class Q8 {
2     public static void main(String[] args) {
3         int n = 1000;
4         int[] array = new int[n];
5         for (int i = 0; i < n; i++) {
6             array[i] = (int) Math.round(Math.random() * 255);
7         }
8         int[] counts = new int[255];
9         for (int i = 0; i < n; i++) {
10
11         }
12         for (int i = 0; i < counts.length; i++) {
13             System.out.println(i + ": " + counts[i]);
14         }
15     }
16 }
```

The program has one mistake. Point out the line number of the mistake, and correct it.

Line 8, “int[] counts = new int[255]” to “int[] counts = new int[256]”

Complete the missing code at line 10.

“counts[array[i]]++;” or “counts[array[i]]+=1;” or “counts[array[i]]=counts[array[i]]+1;”

9. (15 points) Read the following Java program, and write down your answer(s).

```
1 public class Q9 {
2     public static int f(int[] a, int n) {
3         int i1 = 0;
4         int i2 = a.length;
5         int c = 0;
6         while (true) {
7             c++;
8             int i = (i1 + i2) / 2;
9             if (a[i] == n) return c;
10            if (c > a.length) return -1;
11            if (a[i] < n) i1 = i;
12            if (a[i] > n) i2 = i;
13        }
14    }
15    public static void main(String[] args) {
16        int n = Integer.parseInt(args[0]);
17        int[] array = new int[100];
18        for (int i = 0; i < array.length; i++) array[i] = i;
19        System.out.println("c = " + f(array, n));
20    }
21 }
```

What is the output of the program when running “java Q9 31”?

c = 4

What is the output of the program when running “java Q9 128”?

c = -1

For command-line input integer numbers between 0 and 99, what is the largest output value of “c” (at line 19)?

7